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G H Raisoni International Skill Tech University, Pune

S.Y B. Tech (Sem-IV)

CAE III- Summer 2026 (2024 Pattern)

TRANSFORM AND NUMERICAL METHODS (25BTBSL401)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) Solve Q1 or Q2 and Q3 and Q4 ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Illustrate the Laplace Transform of 1) $\sin^2 t$ 2) $e^{4t}\cos 3t$	[5]	1	2
b)	Evaluate the Laplace Transform for function $f(t) = t\,e^{-3t}\sin 2t$	[5]	1	3
	OR			
Q 2.(a)	Illustrate the Laplace Transform of $e^{2t}\cos 2t \sin 3t$	[5]	1	2
(b)	Evaluate the integral using Laplace Transform $\int_0^\infty t\,e^{-2t}\cos 2t\ dt$.	[5]	1	3
Q 3.(a)	Illustrate the Z Transform of i) $2^k + 3^k$, $k \geq 0$ ii) $e^{-3k}\cos 4k$, $k \geq 0$	[5]	3	2
(b)	Obtain f (k), given that $12f(k+2) - 7f(k+1) + f(k) = 0,\ k \geq 0, f(0) = 0, f(1) = 3.$	[5]	3	3
	OR			
Q. 4(a)	Illustarate the Z Transform of $k\,5^k$ $k \geq 0$	[5]	3	2
(b)	Solve the inverse Z Transform of $\frac{z}{(z-1)(z-2)}$ for $\lvert z\rvert \geq 2$.	[5]	3	3